

EasyNet 状态机器现实审计与修正版

当前代码事实、RFC-005 目标契约与未收敛缺口

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Abstract

本文修正早期只描述 clean target 的状态机草稿。本文把当前代码已经实现的事实、RFC-005 的目标契约、仍未收敛的缺口分开陈述, 覆盖 join/start、namespace resolve、invoke、Backend HTTP read model、Frontend typed state、runtimestop 与 reset/self-uninstall。没有源文件证据的状态不会被写成已实现。

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1 更正结论

现实审计规则

前一版文档的问题是把目标架构、当前代码和缺口混在一起。本文要求每个状态机同时给出 `Currentimplementation`、`Targetcontract`、`Gap`。没有源文件证据的状态只能写成目标，不能写成当前事实。

核心结论如下：

- `JoinConnectionState` 已经存在，但状态码不是一一对应，不足以定位所有中断状态。
- `runtimestart` 已经在 `credential verify`、`Hub session endpoint`、`daemon boot failure` 上记录 `snapshot`，且 `join autostart` 失败会显式报错。
- `runtimestop` 当前是命令阶段对象，不是持久化产品状态机。
- `reset` 当前是 `local unpair/reset`，不是完整 `self uninstall/leave`。
- `Backend prepare/submit` 已基本切到 `resolver-selected route`；`read-model`、`media`、`download`、`history` 仍有迁移缺口。
- `Axon resolver negative reason` 只能使用 `proto/RFC-005` 固定集合，`runtime` 文本应进入 `failure locator detail`。

2 建模规则

状态机	Owner	当前完整度	说明
Join credential/runtime admission	CLI	部分完整	有 <code>snapshot</code> ， <code>code</code> 粒度不足
Runtime start/session/presence	CLI daemon + Hub	部分完整	需要更强 <code>admission evidence</code>
Namespace resolve	Axon + CLI daemon	协议完整，产品未全收敛	<code>ResolveAnswer</code> 已规范
Invocation/receipt	Axon + Backend + CLI	部分完整	<code>producer failure receipt</code> 未全覆盖
Backend HTTP read model	Backend	部分完整	<code>pages</code> 局部迁移，仍有字符串解析
Frontend connected/degraded	Frontend	部分完整	<code>invoke route-aware</code> ， <code>media</code> 仍解析字符串
Runtime stop	CLI	缺产品状态机	<code>StopPlan</code> 不是持久化状态
Reset/self uninstall	CLI + Backend/Hub	缺失	<code>reset</code> 不等于 <code>uninstall</code>

3 当前代码审计

- **Join snapshot:** `EasyNet-Cli/src/runtime/join_connection_state.rs`。当前已定义 `JoinConnectionState`、`JoinTransition`、`JoinFailureCode`、`JoinConnectionSnapshot`；缺口是多个状态共用同一 `code`。
- **Join autostart:** `EasyNet-Cli/src/facade/cli/join.rs`。 `pairing accepted` 后默认 `start daemon`；`start` 失败显式报错。

- **Runtime start:** EasyNet-Cli/src/facade/cli/start.rs. credential verify、Hub endpoint probe、daemon boot failure 都记录 snapshot; 仍需保证 online 只在 session admission evidence 后记录。
- **Stop:** EasyNet-Cli/src/facade/cli/stop.rs. 当前有 StopPlan/StopStage; 缺少持久化 stop state/code。
- **Reset:** EasyNet-Cli/src/facade/cli/reset.rs. active runtime 时拒绝 reset; best-effort revoke 后删除本地 credentials; 当前不是 self uninstall。
- **Backend route facade:** EasyNet/backend/internal/logic/ability/resolve_invoke_route.go. 已经调用 daemon resolver; negative 转 typed validation error。
- **Backend prepare/submit:** prepareEnvelopeLogic.go, submitSignedInvocationLogic.go. prepare 使用 selected route; submit 校验 prepare-time route binding。
- **Backend pages:** listPagesLogic.go, helpers.go. pages unavailable DTO 局部存在; 仍通过错误字符串判断 offline/not advertised。
- **Frontend invoke DTO:** Frontend/src/lib/api/easynet-abilities.ts. target_ura/tool_name 已存在, failure locator 类型存在。
- **Frontend media:** Frontend/src/store/media-channel-*.ts. 仍匹配 PresenceRegistry 字符串, 需要 typed failure。
- **macOS client:** EasyNet-Client/.../DeviceConnectionManager.swift. 仍可能合成 ability_ura; 应删除 canonical URA synthesis。
- **Axon namespace proto:** EasyNet-Axon/core/proto/axon/v1/namespace.proto. ResolveAnswerKind 和 NegativeReason 已规范。
- **Axon receipt proto:** EasyNet-Axon/core/proto/axon/v1/invoke.proto. InvocationReceipt.failure 已存在; producer 覆盖不足。

4 Join / Runtime Start 状态机

4.1 当前实现状态

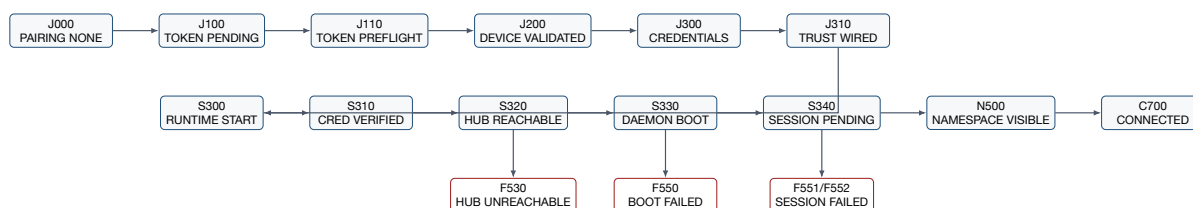
Current state	Code	Wire state	问题
PairingNone	J000	PAIRING_NONE	OK
PairingTokenPending	J100	AUTH_TOKEN_READY	与 preflight/consumed 混码
PairingTokenPreflighted	J100	AUTH_TOKEN_READY	与 pending 混码
PairingTokenExpired	F510	JOIN_REJECTED	OK
PairingTokenConsumed	J100	JOIN_REQUESTED	不应与 pending 混码
DeviceValidatedJoining	J200	PAIRING_ACCEPTED	OK
CredentialsSaved	J300	HUB_PREFLIGHT	与 trust/preflight 混码
LocalTrustWired	J300	HUB_PREFLIGHT	与 credential 混码
RuntimeStarting	J400	DAEMON_BOOT	与 daemon booting 混码
HubCredentialVerified	J300	HUB_PREFLIGHT	与本地保存混码
HubSessionEndpointReachable	J300	HUB_PREFLIGHT	与 credential 混码
DaemonBooting	J400	DAEMON_BOOT	与 runtime starting 混码
SelfSessionAdmissionPending	J500	SESSION_CONNECTING	OK
ConnectedOnline	J800	FRONTEND_CONNECTED	与 suspect/draining 混码

ConnectedSuspect	J800	DEGRADED	不应与 online 混码
ConnectedDraining	J800	OFFLINE	不应与 online 混码
DisconnectedRemoved	F530	OFFLINE	与 unknown 混码
ConnectionUnknown	F530	OFFLINE	与 removed 混码
Failed	F000	FAILED	应减少使用

4.2 目标状态码

Code	Target state	Transition in	Surface
J000	PAIRING_NONE	none	doctor
J100	PAIRING_TOKEN_PENDING	T01_CREATE_PAIRING	frontend
J110	PAIRING_TOKEN_PREFLIGHTED	T02_PREFLIGHT_TOKEN	CLI
F120	PAIRING_TOKEN_EXPIRED	token validation	CLI/frontend
F130	PAIRING_TOKEN_CONSUMED	token validation	CLI/frontend
J200	DEVICE_VALIDATED_JOINING	T03_VALIDATE_TOKEN	CLI
J300	CREDENTIALS_SAVED	T04_SAVE_CREDENTIALS	doctor
J310	LOCAL_TRUST_WIRED	T05_WIRE_LOCAL_TRUST	doctor
S300	RUNTIME_STARTING	T08_BOOT_DAEMON	runtime status
S310	HUB_CREDENTIAL_VERIFIED	T06_VERIFY_CREDENTIAL	doctor
S320	HUB_SESSION_ENDPOINT_REACHABLE	T07_CONNECT_SESSION_ENDPOINT	doctor
S330	DAEMON_BOOTING	T08_BOOT_DAEMON	runtime status
S340	SELF_SESSION_ADMISSION_PENDING	T09_OPEN_SELF_SESSION	daemon status
N500	NAMESPACE_VISIBLE	T10_ADMIT_PRESENCE	backend
C700	CONNECTED_ONLINE	T11_REFETCH_READ_MODEL	frontend
C710	CONNECTED_SUSPECT	degraded event	frontend
C720	CONNECTED_DRAINING	T12_REMOVE_PRESENCE	frontend
C790	DISCONNECTED_REMOVED	presence removed	frontend
C791	CONNECTION_UNKNOWN	missing evidence	frontend

4.3 状态图



4.4 Failure 修正

Code	Failure	Interrupted transition	Retryable
F500	JOIN_FAILED_PREFLIGHT	T02_PREFLIGHT_TOKEN	maybe
F510	JOIN_FAILED_VALIDATE	T03_VALIDATE_TOKEN	no/maybe
F520	START_FAILED_CREDENTIAL_VERIFY	T06_VERIFY_CREDENTIAL	maybe
F530	HUB_UNREACHABLE	T07_CONNECT_SESSION_ENDPOINT	yes
F550	DAEMON_BOOT_FAILED	T08_BOOT_DAEMON	yes/maybe

F551	SELF_SESSION_ADMISSION_FAILED	T09/T10	maybe
F552	SESSION_RUNTIME_UNAVAILABLE	T09_OPEN_SELF_SESSION	yes
F560	RESOLVE_UNAVAILABLE	T11_REFETCH_READ_MODEL	yes

缺口

用户看到的 dendrite bridge library not found 不应再粗略归入 DAEMON_BOOT_FAILED; 目标应投影成 F552 SESSION_RUNTIME_UNAVAILABLE, interrupted transition 为 T09_OPEN_SELF_SESSION。

5 Namespace Resolve 状态机

Axon 的合法 answer kind 是 NON_DISPATCHABLE、DELEGATION、FINAL_ROUTE、NEGATIVE。合法 negative reason 是 UNSPECIFIED、NXDOMAIN、NODATA、NOROUTE、STALE、UNAUTHORIZED、THROTTLED、OVERLOADED、REFUSED、LOOP。

Resolve state	Dispatch allowed	Required output
FINAL_ROUTE	yes	selected ability_ura, route_ura, dispatch_name, next_hop, evidence
DELEGATION	no local dispatch	peer hub or delegated authority
NEGATIVE	no	negative reason, canonical failure code, locator
NON_DISPATCHABLE	no	catalog/resource metadata only
RESOLVE_UNAVAILABLE	no	source, stage, retryable, message

6 Invoke 状态机

通用路径是:

```
User action or CLI command
-> ResolveQuery
-> ResolveAnswer.FinalRoute
-> prepare canonical envelope
-> client signs
-> submit
-> admission
-> dispatch resolver-selected next_hop
-> exactly one terminal InvocationReceipt
```

Variant	Expected next hop	Extra failure cases
同用户同 realm 同设备	LocalDeviceAbility 或 daemon local	ability absent, local execution failed
同用户同 realm 其他设备	hub/device session route	PresenceRegistry miss, stale route
同用户跨 realm	delegation then peer final route	realm not found, delegation refused

跨用户同 realm	policy-authorized	policy denied/hidden,
跨用户跨 realm	hosted/device route	owner not found
	peer delegation + peer policy	delegation refused,
		unauthorized
Hosted agent ability	HostedAgentViaDevice	agent not advertised, host
		device offline
Hub/local ability	LocalHubAbility	hub overloaded, ability not
		found
Streaming/download	final route + stream contract	body committed before
		failure, trailer needed
Terminal/resource	final route + receipt	producer must fill Invocation
producer		Receipt.failure

6.1 Canonical failure code

RFC-005 主分类必须使用: REALM_NOT_FOUND, REALM_DELEGATION_REFUSED, ALIAS_LOOP, ALIAS_AUTHORITY_INVALID, TARGET_NOT_FOUND, OWNER_NOT_FOUND, ABILITY_NOT_FOUND, ABILITY_OWNER_SHADOWED, ABILITY_QUERY_CONFLICT, ROUTE_NOT_FOUND, ROUTE_STALE, RESOURCE_NOT_FOUND, RESOURCE_SCOPE_DENIED, POLICY_DENIED, POLICY_HIDDEN, CAPACITY_THROTTLED, EXECUTION_OVERLOADED, NO_DISPATCHABLE_NEXT_HOP, RESOLVE_R_PROFILE_NOT_AUTHORITATIVE, EXECUTION_FAILED。

7 Backend HTTP 与 Frontend 状态

Situation	Correct HTTP shape	Product meaning
DB rows exist, resolver unavailable	200 + rows + resolve_unavailable[]	stale/degraded read model
route negative	typed 400/422	action invalid now
daemon/bridge unavailable	typed 503	retryable runtime
		unavailable
auth/policy denied	typed 403	permission problem
backend invariant violation	typed 500	backend bug

Frontend 只能从 Backend typed DTO、resolver catalog route、receipt failure、download trailer/control frame 渲染状态。不能从空列表或 human string 推断 connected/offline。

8 Runtime Stop 状态机

当前 stop.rs 已有较好的命令对象分层: StopShape, StopPlan, StopStage。缺口是它没有持久化状态码, 所以 doctor 无法回答“stop 停在哪个 transition”。

Code	Target state	Current stage
O000	STOP_REQUESTED	command start
O100	STOP_SHAPE_CLASSIFIED	StopPlan::from_state
O200	REVOKE_PRESENCE	stage_revoke
O300	STOP_HEARTBEAT	stage_stop_heartbeat
O400	STOP_DAEMON	stage_stop_daemon
O500	SWEEP_DAEMONS	stage_sweep_daemons
O600	STOP_AXON_RUNTIME	stage_stop_axon_runtime

O700	CLEANUP_RUNTIME_STATE	stage_cleanup_state
O900	STOPPED	success

9 Reset 与 Self Uninstall

当前 `reset.rs` 是 local reset/unpair。完整 self uninstall/leave 应拆成独立语义:

Command	Meaning	Failure policy
<code>easynet reset</code>	local credential/runtime cleanup	可继续, 但必须记录 partial
<code>easynet self uninstall/leave</code>	authoritative Hub-side leave + local cleanup	remote failure 必须 terminal 或 explicit partial

目标 self uninstall 状态:

```
U000 UNINSTALL_REQUESTED
-> U100 RUNTIME_GUARD
-> U200 REVOKE_SESSION
-> U300 REMOVE_HUB_MEMBERSHIP
-> U400 REMOVE_LOCAL_TRUST
-> U500 DELETE_CREDENTIALS
-> U600 CLEANUP_RUNTIME_FILES
-> U900 UNINSTALLED
```

10 RFC-005 计划对齐审计

本节修正一个关键问题: 状态机器不能只写 clean target。必须和 Axon 仓库已经固定的 RFC-005 计划包以及当前代码对齐。规范来源包括 005-ura-namespace-resolution-dns-plan.md 以及 00-intent.md 到 06-decisions-log.md。结论是: Axon 中 authoritative resolver 状态机器已经存在, 产品仓库必须向它收敛; CLI/Backend/Client 不能再自建 route inference。

10.1 Axon 当前代码事实

Gate	Code evidence	Result
A1 typed owner parser	EasyNet-Cli/src/ura.rs re-export easynet_axon::ura::*; AbilitySelector 只做投影。	CLI 方向正确。
A2 proto	core/proto/axon/v1/namespace.proto 定义 NegativeReason, ResolveAnswerKind, NextHop, ResolveAnswer.	Axon proto 已有。
A4 resolver object	core/runtime-rs/src/services/namespace/resolver.rs 定义 NamespaceRouteResolver 和 V1Resolver.	Axon runtime 已有对象边界。
A6 invoke consume	core/runtime-rs/src/services/invoke/resolver_consume.rs 消费 ResolveAnswer; rpc_handlers.rs 分支处理 FinalRoute/Delegation/Negative/NonDispatchable.	带 easynet.target_ura 的 invoke 已接入。

A7 release profile	profile.rs 从 Preview 起步; resolver_cons ume.rs 对非 product-dispatchable profile 机械拒绝 dispatch/forward。	已实现 fail-closed gate。
Receipt failure	invoke.proto 有 InvocationReceipt.failure =30; resolver negative 可映射到带 namespace evidence 的 pb::Error。	协议载体已有; producer 覆盖仍需审计。

因此 product invoke 必须包含这个 pre-dispatch gate:

```
I120 RESOLVE_ROUTE
-> I130 CHECK_RESOLVER_PROFILE
    Preview/ShadowRead + FinalRoute/Delegation
    -> I902 NO_DISPATCHABLE_NEXT_HOP
    AuthoritativeLocal/Production + FinalRoute
    -> I300 LOCAL_DISPATCH
    AuthoritativeLocal/Production + Delegation
    -> I400 PEER_DELEGATION
    Negative
    -> I900 TERMINAL_FAILURE
    NonDispatchable
    -> I902 NO_DISPATCHABLE_NEXT_HOP
```

10.2 当前跨仓库未收敛点

Repo	Observed code	RFC-005 risk
EasyNet-Cli	src/eal/interpreter.rs 仍用 TargetOwn edAbilityUra::from_selector 后调用 federation forward。	EAL child invoke 可能绕过 resolver-selected route。
EasyNet-Cli	src/runtime/agents/invoke_ability.rs input schema 仍要求 ability_ura。	只能作为 internal/diagnostic; 产品路径不能要求 naked ability URA。
EasyNet-Cli	src/runtime/failure_codes.rs 仍按 PresenceRegistry/not advertised 文本分类。	failure projection 仍不是完全 canonical。
Backend	backend/internal/runtime/pty_driver.go 构造 axon.PublishedAbilityURA(calle eURA,ability)。	PTY 路径仍在 Backend 侧 合成 device-owned ability URA。
Backend	backend/internal/logic/page/helpers.go 用字符串判断 pages host offline。	pages failure 还未完全 typed negative。
Backend	backend/internal/handler/file/pumps.go 已写 X-EasyNet-Failure trailer。	后端侧部分完成; browser/client 解码仍需核 验。
Frontend	media-channel-invocation.ts 和 media-c hannel-store.ts 仍匹配 notinPresenceR egistry。	UI 仍消费 transport 文本而 不是 FailureLocator。
EasyNet-Client	macOS 旧函数名 Ability.synthesizeA bilityURI 生成 /agents/.../abilities/ ...@version。	与 RFC-005 最终 URA grammar 不兼容。

10.3 修正后的 invoke 状态族

不能再用一个泛化 invoke 图隐藏 route class。正确模型是公共前缀加显式分支：

```
COMMON
I000 REQUEST_RECEIVED
I020 VALIDATE_TARGET_URA_AND_TOOL_NAME
I040 LOAD_CALLER_SUBJECT_AUTHORITY
I080 BUILD_RESOLVE_QUERY
I120 RESOLVE_ROUTE
I130 CHECK_RESOLVER_PROFILE

SAME REALM FINAL ROUTE
I300 DISPATCH_LOCAL_OR_REMOTE_SELECTED_ROUTE
I500 EMIT_TERMINAL_RECEIPT
I900 COMPLETED

CROSS REALM DELEGATION
I400 FORWARD_TO_PEER_HUB
I420 PEER_RESOLVE_AUTHORITATIVELY
I440 PEER_FINAL_ROUTE_OR_NEGATIVE
I500 RETURN_RECEIPT_CHAIN
I900 COMPLETED

NEGATIVE / NON-DISPATCHABLE
I700 MAP_NEGATIVE_TO_FAILURE_CODE
I720 ATTACH_FAILURE_LOCATOR
I740 WRITE_INVOCATION_RECEIPT_FAILURE
I950 FAILED_TYPED
```

Condition	Receipt code	Locator layer
unknown realm	REALM_NOT_FOUND	Realm
delegation refused	REALM_DELEGATION_REFUSED	Authority
target absent	TARGET_NOT_FOUND	Identity
owner absent	OWNER_NOT_FOUND	Owner
ability absent with owner present	ABILITY_NOT_FOUND	Ability
full ability URA plus secondary	ABILITY_QUERY_CONFLICT	Ability
ability name		
no route	ROUTE_NOT_FOUND	Route
stale route lease	ROUTE_STALE	Route
positive but not dispatchable	NO_DISPATCHABLE_NEXT_HOP	Route
resource absent	RESOURCE_NOT_FOUND	Resource
resource scope denied	RESOURCE_SCOPE_DENIED	Resource/Policy
policy denied or hidden	POLICY_DENIED / POLICY_HIDDEN	Policy
	EN	
resolve capacity exhausted	THROTTLED	Route
execution plane saturated	OVERLOADED	Execution

10.4 Join/Stop/Uninstall 修正

Join 必须把 credential accepted 和 runtime connected 拆开：

```
J000 TOKEN_RECEIVED
J050 PAIRING_PREFLIGHT
J100 PAIRING_ACCEPTED
```

```
J150 CREDENTIALS_PERSISTED
J200 BOOT_REQUESTED
J230 BRIDGE_LIB_VALIDATED_OR_NOT_REQUIRED
J260 HUB_SESSION_ENDPOINT_REACHABLE
J300 DAEMON_PROCESS_STARTED
J360 AXON_RUNTIME_READY
J420 SELF_SESSION_ADMITTED
J500 PRESENCE_ADVERTISED
J700 BACKEND_READ_MODEL_VISIBLE
J900 CONNECTED
```

Stop 至少持久化:

```
S000 STOP_REQUESTED
S100 SIGNAL_DAEMON
S200 DRAIN_SESSIONS
S300 UNPUBLISH_PRESENCE
S400 STOP_RUNTIME
S500 VERIFY_SOCKET_CLOSED
S900 STOPPED
S950 PARTIAL_STOP
```

Self uninstall/leave 至少持久化:

```
U000 UNINSTALL_REQUESTED
U100 RUNTIME_GUARD
U200 REVOKE_SESSION
U300 REMOVE_HUB_MEMBERSHIP
U400 UNPUBLISH_DEVICE_AND_ABILITIES
U500 REMOVE_LOCAL_TRUST
U600 DELETE_CREDENTIALS
U700 CLEAN_RUNTIME_FILES
U900 UNINSTALLED
U950 PARTIAL_UNINSTALL
```

11 修复计划

11.1 EasyNet-Cli

- 将 `JoinConnectionState::code()` 改成一一对应。
- 拆分 boot failure、self-session admission failure、session runtime unavailable。
- 确保 `ConnectedOnline` 只在 self-session admission/PresenceRegistry evidence 后记录。
- 为 stop/reset 增加持久化 lifecycle snapshot。
- 所有 ability/resource/meta 调用走 resolver-selected route。
- terminal/download/resource producer 统一写 `InvocationReceipt.failure`。

11.2 EasyNet Backend

- 保持 `resolveSelectedInvokeRouteFrom` 为 prepare route authority。
- 审计所有直接要求 `ability_ura` 的 handler, 改为 catalog route 或 `target_ura+tool_name`。
- 删除 `page/media/read-model` 字符串错误解析, 改 `typed failure projection`。

- read model 返回 degraded DTO; action 返回 typed 4xx/503。

11.3 EasyNet Frontend / Client

- 删除 canonical ability URA synthesis。
- 所有 action button 绑定 resolver catalog route。
- 用 typed failure locator 替代 PresenceRegistry 字符串匹配。
- download client 解码 X-EasyNet-Failure trailer/control frame。

11.4 EasyNet-Axon

- 保持 NegativeReason 与 RFC-005/proto 一致。
- 暴露 runtime/admission/transport error 到 canonical failure code + locator 的 mapper。
- 所有 non-success terminal receipt 必须填 InvocationReceipt.failure。

12 验收标准

- easynet join --boot yes 在 daemon 无法 reach/admit runtime 时失败, 且 snapshot 标出 interrupted transition。
- easynet doctor --json 显示 machine、state code、transition、failure locator、next action。
- runtimestop 和 reset/self-uninstall 有稳定状态码。
- /api/v1/pages 不因 agent-not-advertised 返回普通 500。
- /api/v1/invocations/history/list 不再因 route-aware UI 请求报 ability_uraisrequired。
- meta.list_resources 有 canonical route 或 typed failure。
- Frontend/Client 不再构造 canonical ability URA。
- browser/download 能读 typed failure trailer。
- failed/timeout/cancelled invocation 都有 InvocationReceipt.failure。